Michael Baffour Awuah

Ph.D. Candidate in Microbiology & Bacteriophage Biology

Department of Biology, Texas A&M University, College Station, TX

baffour@tamu.edu (979) 326-6958 mbaffour LinkedIn

BACTERIOPHAGE BIOLOGY | CRISPR ENGINEERING | FUNCTIONAL GENOMICS | RNA-SEQ | MOLECULAR CLONING | PROTEIN BIOCHEMISTRY | BIOINFORMATICS

Education

Ph.D., Biology (Microbiology)

Aug 2022 – Present

Texas A&M University, College Station, TX

- **Dissertation:** Delaying the Inevitable – Lysis Inhibition in Bacteriophage N4
- **Advisor:** Dr. Jolene Ramsey

B.Sc., Biochemistry (Second Class Upper Division)

Aug 2016 – Jun 2020

University of Cape Coast, Cape Coast, Ghana

- **Dissertation:** Physicochemical, Functional and Nutritional Analysis of a Juice Blend and Mixed Fruit Nectar Formulated From Pineapple and Underutilized Tropical Fruits
- **Mini Project:** Qualitative Phytochemical Analysis, Total Antioxidant Activity and Total Phenolics of *Terminalia catappa*
- **Advisor:** Dr. Jerry Ampofo-Asiama

Research Experience

Department of Biology, Texas A&M University

Aug 2022 – Present

Graduate Researcher, Ramsey Laboratory

College Station, TX

- Identified and characterized the minimal gene set required for Bacteriophage N4 lysis through systematic CRISPR-based deletion analysis and functional genomics, discovering a novel lysis regulator.
- Performed RNA-seq transcriptomic profiling to uncover transcriptional differences between wild-type N4 and rapid-lysis lineages, revealing key regulatory networks governing lysis timing.
- Engineered bacteriophage N4 mutants using combinatorial genetics, designing complex genetic circuits to study how bacterial growth phases affect translation machinery availability during infection.
- Isolated, purified, and characterized three novel *E. coli* 4s-infecting bacteriophages from environmental samples using TEM, restriction analysis, and lysis phenotype assays.
- Quantified LPS/endotoxin content and total protein in phage lysates via BCA assays, comparing disruptin vs. spanin LPS-quenching properties.

Show more

- Sequenced, assembled, and annotated genomes of three novel phages, integrating morphological, genomic, and lysis kinetics data to classify phage diversity.
- Designed and optimized primers for molecular cloning, including plasmid design, construction, and verification for gene expression and complementation studies using SnapGene.
- Developed fluorophore-based reporter assays (GFP/mCherry) for real-time monitoring of lysis protein expression.

West African Center for Cell Biology of Infectious Pathogens

Jun – Jul 2022

Graduate Intern, Molecular Biology Laboratory

University of Ghana, Legon, Ghana

- Performed antimicrobial susceptibility testing on sixteen drug-resistant bacterial strains from the Greater Accra Regional Hospital, providing data-driven disinfectant recommendations for infection control.
- Determined MICs of three commonly used disinfectants against bacteria isolated from nose masks, contributing to public health research.

Department of Biochemistry, University of Cape Coast

Sep 2020 – Oct 2021

Research & Teaching Assistant

Cape Coast, Ghana

- Conducted molecular characterization of *H. pylori* virulence and antibiotic resistance genes from clinical gastroduodenal specimens, linking genotypes to pathology severity.
- Investigated qualitative and quantitative phytochemical properties of methanolic leaf extracts of *Anthocleista nobilis*, its extraction kinetics, and antimicrobial susceptibility against selected bacterial strains.
- Assisted in optimizing ultrasound-assisted extraction of phenolic compounds and antioxidant activity from *Solanum torvum*.
- Served as field and laboratory assistant for MPhil Fisheries Science studies on genetic diversity and microbiome of small pelagic fish species and nutritional quality of oysters in Ghana.

Show more

- Mentored third-year undergraduate Biochemistry students in a study on the antimicrobial activity of ginger and phytochemical properties of *Garcinia kola*.
- Guided final-year Clinical Nutrition and Dietetics students in research on the antioxidant properties of *Parkia biglobosa*.
- Planned experiments, prepared reagents, ran trials, organized experimental setups, and guided students in lab sessions to ensure accurate and reproducible results.
- Managed an inventory of chemicals and equipment in the laboratory and ensured equipment and glassware were maintained in optimal working condition.

Department of Biochemistry, University of Cape Coast

Jan 2018 – Jun 2020

Undergraduate Researcher

Cape Coast, Ghana

- Deduced phytochemical composition of almond pulp extracts and elucidated antioxidant activity through DPPH radical scavenging activity and total phenolic content via chemical and spectrophotometric methods.
- Optimized mixing ratios and formulated juice blends and mixed fruit nectars from underutilized tropical fruits (pineapple, pawpaw, soursop).
- Studied the effects of blending and blending ratios on the nutritional quality of the product.
- Investigated changes in physicochemical properties — color, pH, titratable acidity, total soluble solids, and viscosity — resulting from blending and nectar formulation.

Noguchi Memorial Institute for Medical Research

Jun – Jul 2019

Research Intern

University of Ghana, Accra, Ghana

- Completed rotations in general parasitology, entomology, molecular biology, vector, and parasite culture laboratories.
- Executed techniques including DNA & RNA extraction, PCR, gel electrophoresis, urine filtration, urine chemistry, microscopy for malaria diagnosis, and Kato-Katz analysis for *Schistosoma* egg identification.
- Cultured *Plasmodium*, *Leishmania*, and *Trypanosoma* spp. to assist in ongoing research across multiple parasitology laboratories.
- Established vector breeding colonies for transmission studies at the Vestergaard Vector Laboratories.

Other Work Experience

The Community Hospital – Ashongman

Jun – Aug 2018

Biomedical Scientist Intern

Accra, Ghana

- Managed laboratory decontamination procedures, monitored equipment performance, and documented data.
- Handled sample intake, assigned pathology numbers, and distributed samples for analysis.
- Prepared reagents and conducted routine diagnostic tests.
- Established and managed a mobile laboratory unit during high-demand periods.
- Conducted phlebotomy and performed cross-matching assays for blood transfusion compatibility.
- Documented test results to support medical diagnosis.

Taifa Polyclinic Jun – Aug 2017

Biomedical Scientist Intern

Accra, Ghana

- Managed workspace decontamination, equipment monitoring, and data documentation.
- Handled sample acceptance, allocation of pathology numbers, and distribution for analysis.
- Prepared reagents for tests and conducted routine diagnostic tests.
- Conducted phlebotomy for blood sample collection.
- Documented test results for medical diagnosis support.
- Administered client feedback questionnaires.

Publications

Awuah MB†, Martin C†, Chamblee JS, Tomaszewski AJ, Sullivan TE, Emilia Q, Tran S, Snowden JH, Niemiec KA, Zhu J, Ramsey J. *Phage N4 uses a SAR endolysin-holin system for host cell lysis.* Journal of Bacteriology. 2026. doi: 10.1128/jb.00538-25. Preprint: bioRxiv 2025, doi: 10.1101/2025.11.12.688109. †Equal contribution.

Debrah MA, **Awuah MB**, Koh A, Ramsey J. *Complete genome sequence of Escherichia Siphophage Serwaa.* Microbiology Resource Announcements. 2026;e01222-25. doi: 10.1128/mra.01222-25.

Presentations

2025

- "Phage N4 lysis inhibition – the viral death race" — Students' Postdoc Research Conference, College Station, TX **Poster**

2024

- "Breaking Up The Breakdown: How An Open Reading Frame Inhibits Lysis In Bacteriophage N4" — Texas ASM Branch Meeting, UTMB, Galveston, TX **Poster**
- "Breaking Up The Breakdown: How An Open Reading Frame Inhibits Lysis In Bacteriophage N4" — Mol. Genetics of Bacteria & Phages Meeting, UW-Madison **Poster**
- "Bacteriophage N4 Lysis Inhibition Unraveling Ancient Mechanism With Modern Insights" — Bio & Chem Sci Symposium, College Station, TX **Oral**
- "Bacteriophage N4 Lysis Inhibition – Ancient Conundrum Meets Novel Culprit" — Texas ASM Branch Meeting, Cedar Hill, TX **Oral**
- "Phage N4 lysis inhibition – the viral death race" — Students' Postdoc Research Conference, College Station, TX **Poster + Blitz**
- "Delaying the Inevitable – Bacteriophage N4 Lysis Inhibition" — BIOGSA, College Station, TX **Oral**

Earlier presentations

2023

- "Delaying the Inevitable – Bacteriophage N4 Lysis Inhibition" — Mol. Genetics of Bacteria & Phages Meeting, UW-Madison **Poster**
- "Phage Lysis Inhibition is Phruitful Procrastination" — Diversity in Science Symposium, Texas A&M University **Poster**
- "Phage Lysis Inhibition is Phruitful Procrastination" — Texas ASM Branch Meeting, Abilene Christian University **Poster**

2021

- "Mendeley Referencing Tool For Dissertation Writing" — GHABSA, UCC **Lecture**

2020

- "G6PD Deficiency" — GHABSA National Congress, UCC **Poster**

Teaching Experience

COURSE	STUDENTS	TERM	INSTITUTION
BIOL 111 Intro Biology I	48	Fall 2022	Texas A&M
BIOL 112 Intro Biology II	48	Spring 2023	Texas A&M
General Biochemistry Practical	209	Fall 2020	U. Cape Coast
Biochemical Techniques I	150	Fall 2020	U. Cape Coast
Diversity of Living Organisms	Class of 2023	Fall 2020	U. Cape Coast
Biological Oxidation & Bioenergetics	169	Fall 2020	U. Cape Coast
Biostatistics	Class of 2022	Fall 2020	U. Cape Coast
Enzyme Action & Catalysis	169	Fall 2021	U. Cape Coast
Biochemical Techniques (Mini Project)	5	Fall 2021	U. Cape Coast

Facilitated microteaching training for new TAs (2023–present).

Funding & Awards

🏆 **Thomas L. Patterson Graduate Student Fellowship (2025)**
Outstanding research in bacteriophage biology – discovery of a novel lysis regulator in Bacteriophage N4

🎓 **Mol. Genetics of Bacteria & Phages Meeting Registration Waiver (2024)**
$525

🎓 **BIOGSA Engagement Scholarship (2023)**
$250 – Texas A&M Department of Biology

🎓 **Mol. Genetics of Bacteria & Phages Meeting Registration Waiver (2023)**
$500

📚 **COCOBOD CMB Scholarship (2012–2015)**

Mentorship

Graduate Student Mentor — Texas A&M University

Annie Koh
Characterization of three *E. coli* 4s phages

Philip Lotaire
N4 Lysis Inhibition, Host Specificity & Receptor Discovery

Matthew Fan
Mechanism of action of bacteriocins

Teja Vemulapalli
Bacteriophage N4 Lysis Inhibition

Research Mentor — University of Cape Coast

Bridget Ahema Agyemang
Phytochemical eval. of *Garcinia kola*

Bertha Klenam Hamenu
Phytochemical eval. of *Garcinia kola*

Ekow Etwire
Phytochemical eval. of *Garcinia kola*

Samuel Ofosu
Phytochemical eval. of *Garcinia kola*

Emmanuel Luri
Phytochemical eval. of *Garcinia kola*

Christian Okine
Phytochemical eval. of *Garcinia kola*

Leadership, Service & Community

Judge — Texas Junior Science & Humanities Symposium — Jan 2025

Judge — SACNAS Diversity in Science Symposium — Apr 2024

Volunteer — Darwin Day, Texas A&M University — Feb 2024 & 2025

Microteaching Mentor — Biology Dept, TAMU — Aug 2023 & 2024

Invited Speaker — Vizta Edu – Study Abroad Journey — Aug 2023

Editor-in-Chief — GHABSA UCC Chapter — 2019–2020

National Editor — Ghana Biochemistry Students Association — 2019–2020

Co-founder & Editor — FillaHQ.com (UCC campus media) — 2017–2020

Volunteer — Verifie Health — Nov 2018 – Present

Member — NUGS UCC Research & Sponsorship Committee — 2018–2019

Volunteer — UCC SRC Community Outreach — May 2017

Volunteer Teacher — Great Messiah International School — 2015–2016

Technical Expertise

MOLECULAR VIROLOGY & SYNTHETIC BIOLOGY

Phage Isolation & Characterization | Phage Genomics | CRISPR-based Engineering | Lysis Mechanism Elucidation | Phage-Host Interactions | Lysis Timing & Kinetics

MOLECULAR BIOLOGY

PCR / qPCR | Molecular Cloning | DNA/RNA Extraction | Gel Electrophoresis | Genome Assembly | Restriction Digestion | TEM | SDS-PAGE | Western Blotting

PROTEIN BIOCHEMISTRY

BCA / Bradford Assays | LPS / Endotoxin Quantification (LAL, phage lysates) | OM Protein Characterization | Fluorophore Reporters (GFP/mCherry)

MICROBIOLOGY & BIOCHEMISTRY

BSL2 Culture (8+ yrs) | AST / MIC Determination | Phytochemical Profiling | DPPH / Folin-Ciocalteu

GENOMICS & BIOINFORMATICS

NGS / RNA-seq | Comparative Genomics | Phylogenetics | Differential Expression | Functional Annotation | SnapGene

COMPUTATIONAL & DATA SCIENCE

R | R Shiny | Python | Git / GitHub | CLI Development | Reproducible Environments | Regression Testing | Data Visualization | Pipeline Development

MACHINE LEARNING & STRUCTURAL PREDICTION

CNN Training (ResNet-18) | Model Evaluation (leave-one-plate-out) | Integrating Third-Party Models (YOLO-seg) | Image Segmentation | Human-in-the-Loop Annotation | AlphaFold 3 / ipTM Screening | HMMER | MAFFT | IQ-TREE

Training & Certifications

- Academy For Future Faculty, Texas A&M University (Fall 2023)
- IBRO-ARC Workshop on Zebrafish as a Tool for Neuroscience Research in Africa (Nov 2021)
- GHABSA Annual National Congress (Feb 2020)
- Training on Food Value Chain Standards (Nov 2019)
- NMIMR Biosafety Training (Jun 2018)
- AI Fluency Framework & Foundations Certificate
- Strategic Thinking Certificate
- Biomedical Research Investigators and Key Personnel Certification

Professional Membership

American Society for Microbiology (ASM) — January 2021 – Present

References

Letters from Texas A&M University and the University of Cape Coast, including my dissertation advisor, are available on request.

Built in a phage lab, mostly after hours.

© 2026 Michael Baffour Awuah | baffour@tamu.edu